Experimental Design Summary (Laparoscopic task – Peg transfer)

Component	Details
Consent	Written informed consent obtained prior to participation (Appendix A and B).
Eligibility screening	Normal or corrected-to-normal vision; no motor or neurological impairments reported.
Handedness	Edinburgh Handedness Inventory (Oldfield, 1971); laterality quotient recorded ($M = 98.79$, $SD = 5.28$). Inclusion criterion: ≥70 (all participants met criterion).
Video game exposure	Video Game Questionnaire (Bediou et al., 2018).
Group allocation	Three groups: (1) camera rotation, (2) no camera rotation, (3) VMA-only control.
Sessions	Four sessions per participant, spaced 7 days apart.

Laparoscopic task (camera rotation and no camera rotation groups only)

Component	Camera rotation group	No camera rotation group
Instruction	Standardised PowerPoint, video tutorial, and experimenter demonstration of instrument handling	Same
Familiarisation	60 s free manipulation of instruments without contacting objects	Same
Task structure	Forward transfer followed by reverse transfer constitutes one trial	Same
Apparatus	Box trainer with Maryland laparoscopic graspers	Same
Camera manipulation	5 levels: 0°, 22.5°, 45°, −22.5°, −45°	Fixed at 0° across all trials
Trial structure	10 trials per session. All participants begin with a 0° trial. Remaining 9 trials consist of fully randomised presentation of all rotation conditions. Within-	10 trials per session, all at 0° camera angle.

Component	Camera rotation group	No camera rotation group
	session order is treated as non-inferential exposure variability.	
Exposure logic	Each participant receives exposure to all rotation conditions; total exposure is equivalent across participants; within-session variability is treated as noise rather than an inferential target	Not applicable
Time limit	300 s per trial; timing begins at first object contact and ends at completion or timeout	Same
Feedback	30 s remaining warning; corrective instruction provided if required	Same
Performance measures	Completion time, completion status (completed vs censored), and error recordings	Same

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VMA-only control group

Component	Details
Task	Visuomotor adaptation (VMA) task only; no laparoscopic training component
Exposure	Completed VMA task across four sessions
Measures	VMA performance metrics (as defined in separate task section)

Experimental Design Summary (VMR + Generalisation)

Phase	Trials	Targets	Feedback	Structure
Baseline	198	11 targets (18 per target)	132 full-feedback + 66 no-feedback	Pseudorandomised order across 11 directions

Phase	Trials	Targets	Feedback	Structure
Adaptation	110	1 target (0°)	Endpoint feedback with 22.5° CW visuomotor rotation	Single-target adaptation under fixed CW rotation for all participants
Generalisation	66	11 targets × 6 repetitions	No feedback; no rotation	6 cycles × 11 targets; randomised order within each cycle